Continuous Care Companion

AI Clinical Documentation, Unified Longitudinal Health Records & Hereditary Risk Detection at the Point of Care

Problem Statement

1. Fragmented patient history at the point of care.

Patients typically carry an incomplete, scattered set of past reports, imaging and discharge summaries across multiple hospitals and clinics, since there is no single system for a patient to hold their own digital records. During consultation, doctors frequently review only what the patient happens to have on hand — physical printouts, phone photos, or partial digital files — while the rest of the longitudinal record (labs, radiology, prior diagnoses) is missing, delaying and sometimes compromising clinical decision-making. India's Ayushman Bharat Digital Mission (ABHA) is building the national rails for a longitudinal health record, but there is no ready mechanism at the clinic level to pull this data into the doctor's workflow at the moment of consultation.

2. Hereditary cancer risk is not systematically captured or flagged.

During consultations, family history of cancer is rarely captured in a structured way, and non-hereditary but high-recurrence-risk patterns within a family are often missed entirely. Without a consistent mechanism to record and score this risk (family history, personal history, lifestyle and social habits), at-risk patients are not identified early enough for preventive screening or timely diagnosis — a gap doctors want closed but currently have no structured tool to address.

3. Manual clinical documentation burdens doctors and compounds the history gap.

India's healthcare system loses millions of clinical hours because doctors spend more time documenting than delivering care: across ~1.3M doctors, 4+ hours a day go into administrative charting, and 68% report burnout. Inaccurate, delayed records in a multilingual setting lead to medical errors and poor care, and because each visit's notes are typed independently and rarely structured, a patient's own consultation history within the same clinic is often not maintained as a continuous, searchable record either — leaving both sides of the record, hospital-side and clinic-side, equally fragmented.

Proposed Solution

Build on Medlio — our AI clinical documentation platform — and extend it into a Continuous Care Companion with three connected capabilities:

- Unified Longitudinal Record: Medlio is developing a Patient App so patients maintain their own digital records — reports, imaging and discharge summaries — in a single system, ending the need to carry multiple physical files to every consultation. The app integrates with ABHA (and hospital/lab systems) to pull a patient's consented historical data into one digital timeline, auto-attached to the consultation so the doctor sees the complete picture, not just what the patient brought.
- AI-Assisted Hereditary Risk Flag: As the doctor takes history at the start of the consultation, Medlio prompts for and extracts family history, personal history and lifestyle/social-habit data, scores hereditary cancer risk against known clinical patterns, and raises an early flag with a suggested screening/preventive pathway — captured as a structured, reusable data point rather than free text buried in notes.
- AI Clinical Documentation: Medlio's ambient AI listens to the natural doctor–patient conversation (speech-to-text trained on medical terminology and Indian accents at >95% accuracy, with multilingual reasoning across English and local languages) and auto-generates structured SOAP notes and patient summaries in real time, coded against ICD-10 / SNOMED CT and integrated directly into EMR/HMIS — no templates, no clicking, just talking. This cuts documentation time and, because every visit is captured in the same structured format, feeds straight into the same longitudinal record the Patient App maintains.

All three capabilities share Medlio's existing consultation pipeline and medical coding layer, so the Patient App, doctor-side documentation and risk flags all live in one structured, reusable record — immediately usable for referrals, insurance and longitudinal tracking rather than a one-off report.

Expected Impact

- Patients stop carrying multiple physical files; doctors consult with the full historical picture, reducing repeat tests and diagnostic delay.
- Structured capture of family/lifestyle risk enables early, preventive intervention for hereditary-pattern cancers.

- 70% reduction in documentation time and 80% reduction in clinical burnout, freeing doctors for direct patient care instead of manual charting; every visit automatically adds to the same structured record.
- 30% increase in medication adherence and 50% increase in patient treatment understanding, from clear, multilingual after-visit summaries.
- Builds directly on ABHA's national digital health infrastructure, keeping the solution interoperable and scalable across hospitals and clinics.
- Reuses Medlio's existing AI consultation and coding pipeline — fast to prototype, low incremental build cost.